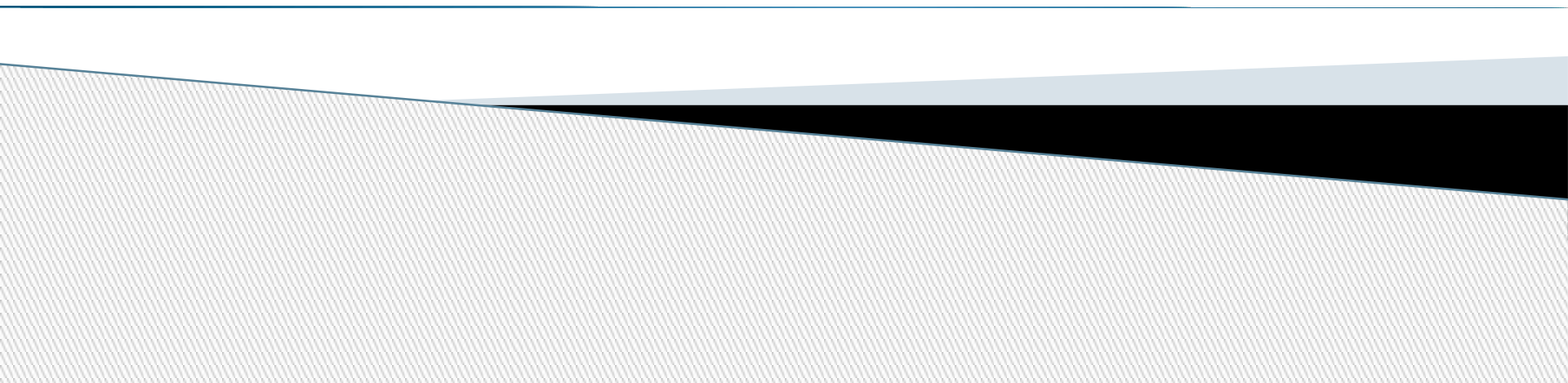


CS 3251

Intermediate Software Design

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CS 3251 Course Philosophy

Good design and programming is not learned by generalities, but by seeing how significant programs can be made clean, easy to read, easy to maintain and modify, human-engineered, efficient, and reliable, by the application of good design and programming practices. Careful study and imitation of good designs and programs significantly improves development skills.

- Kernighan and Plauger

CS 3251 Ground Rules

- Assignments must be submitted on time - **no late work will be accepted**
- Work **must** be your own!!!
- **No** laptops open, texting, tablets or smartphones during exams or quizzes
- You will be called upon periodically to answer questions
- You'll get out of this course exactly what you put into it, so be prepared to work hard
- Be prepared and **on-time** for quizzes. You know when they will be.
- You must take quizzes and exams with your section. Otherwise, zero!
- Make sure to avail yourself of available help, e.g., office hours, TAs, recitation.
- It is **your responsibility** to seek assistance!

CS 3251 - The Socials

- I think of this course as having a “Read Only” social media policy
 - Feel free to use online sources to learn, study and debug
 - You may not “post” anywhere to get help
 - Don’t create questions on StackOverflow
 - Don’t post questions to Quora
 - You may not make any of your code publicly available on Github or elsewhere!!!

Course Information

- Slides will be posted to Brightspace
- All assignments will be coordinated through Github
- All schedule information will be kept at:
<https://vuse-cs3251.github.io/schedule/>

CS 3251 Course Content

- Focus on topics important to developing & maintaining quality software:
 - Modern tools and methods of programming
 - Developing, documenting, testing, & applying reusable classes & object-oriented frameworks using C++
 - Reuse of patterns & software components
- Patterns will be taught to provide good role models for software designs & to clearly articulate design tradeoffs
- It is assumed you know all C++ topics covered in CS 2201

CS 1101 & 2201 - Prev. Knowledge

- Procedural and object-oriented programming
- Data structures, iterators & algorithms
- C/C++ Language
 - Raw & smart pointers
 - Constructors & destructors
 - Error handling with exceptions
 - Maybe *auto* variable declaration
- Programming Skills
 - Test-driven development
 - IDE, compiler & debugger

CS 3251 Course Contents

- The course will cover techniques that address:
 - Source code management
 - Automated testing
 - Decentralized software architecture
 - Architectural and design patterns
 - Class interface definition
 - Component reuse
 - Module decomposition
 - Hierarchical classification
 - Extensible object-oriented application frameworks

Where from here?

- CS 3281 - Operating Systems
 - Threading, coordination primitives
- CS Graphics
- CS Everything!!!
 - Patterns, patterns, patterns

CS 3251 Course Work

- Programming assignments
 - Assignments written in C++
 - Must compile and run using the Clang & GCC compilers and tools
 - Must run on Travis CI service!
 - **Must be done individually**
 - **NO late submissions**
- Programs will be graded based on:
 - 40% Execution correctness
 - 30% Structure (e.g., modularization, information hiding, etc.)
 - 10% Insightful programming (e.g., developing reusable class components, etc.)
 - 10% Consistent style (e.g., capitalization, indenting, etc.)
 - 10% Appropriate commenting style

CS 3251 Course Work

- Quizzes
 - Nearly every week
 - **Must not be late!!!**
- Two Midterms and Final Exam
- The relative weighting of each portion of the course is:
 - 45% Programming projects
 - 25% Quizzes
 - 30% Exams; each 10%

CS 3251 Course Materials

- Books:
 - Highly recommend **C++ Primer (5th Ed.)** by Lippman, et. al.
 - Highly recommend **Design Patterns** by Gamma, Helm, et. al.
 - **Effective Modern C++** by Meyers
 - **The C++ Programming Language** by Stroustrup
- Tools:
 - CLion IDE by JetBrains
 - Clang compiler (built on LLVM), or GCC
 - GDB debugger
 - CMake build tool - already part of CLion

So, Who Am I?

- Dr. Jerry Roth
 - Email: j.roth@vanderbilt.edu
 - Office: FGH 220
 - Phone: (615) 322-2907
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- Office Hours (in FGH 220)
 - Tuesdays 9:30-11:30
 - Tuesdays 1:00-2:00
 - Or any time by appointment

So, Who Am I?



And, who are you?

- Now I want to meet you...
- Name pronounced right.
- Where you're from.
- One interesting fact (brief).